

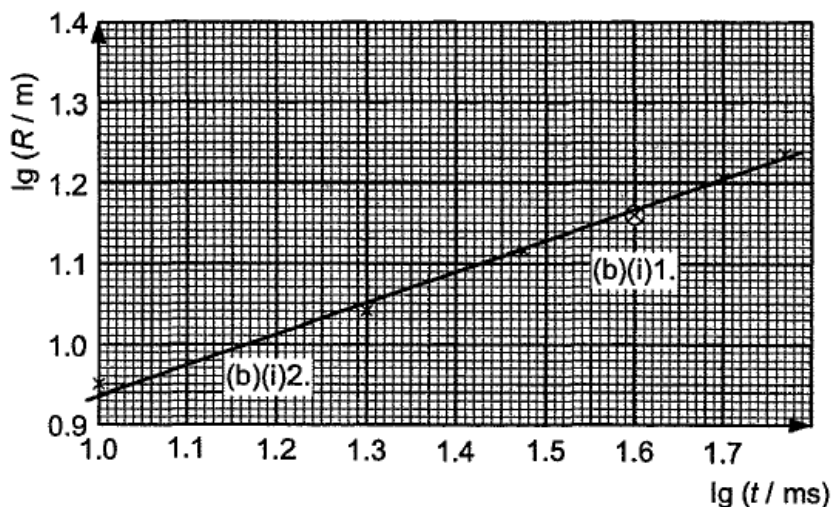
- 7 (a) (i) From $t = 0$ to 2 ms, gradient of the graph is high implies that the initially the rate at which the radius of the fireball increases is very high.
From $t = 2$ ms onwards, gradient of the graph reduces further implies that the rate at which the radius of the fireball increases is reducing further.

(ii) Explosion is hazardous to people because

- for a person standing at a region of 3 m to 5 m from the source, the rate of increase of radius of the fire ball is large.
- even for a person standing at a distance of 12 m from the source, the time taken for the fire ball to reach the person is in ms, which is lesser than the human reaction time.
- people are not able to react within such a short time interval.

Note: Do not misread the time to be in seconds. Many candidates did not refer to the very short time at all and some were concerned with physical damage to the room and suggested that the room would explode.

- (b) (i) At $t = 40$ ms, $R = 14.6$ m.
 $\lg R \approx 1.16$ and $\lg t \approx 1.60$.



(ii) gradient = $\frac{1.16 - 0.96}{1.59 - 1.11} \approx 0.42$

$$(iii) R^n = kt^m$$

$$\rightarrow n \lg R = m \lg t + \lg k$$

$$\rightarrow \lg R = \left(\frac{m}{n}\right) \lg t + \left(\frac{1}{n}\right) \lg k$$

$$\frac{m}{n} = \text{gradient} \approx 0.4$$

Given that n and m are integers, making m as small as possible, implies $n = 5$ and $m = 2$.

Note: A significant number candidates gave answers in non-integer form.

(c)

Radius of fireball, R / m	Volume of container, $V / 10^{-3} \text{ m}^3$	$c = \frac{R^5}{V} / 10^7 \text{ m}^2$
14.6	12.5	5.30
14.0	10.0	5.38
13.1	7.5	5.35
12.2	5.0	5.41
10.6	2.5	5.35
Average		5.36

Conclusion: Since c is a constant, hence the relationship $R^5 = cV$ is true.

(d) (i) One physical quantity on which the constant c will depend is its surrounding pressure.

(ii) Even in an open space, the force of explosion could hurl fragments and objects at high speeds over large distances.